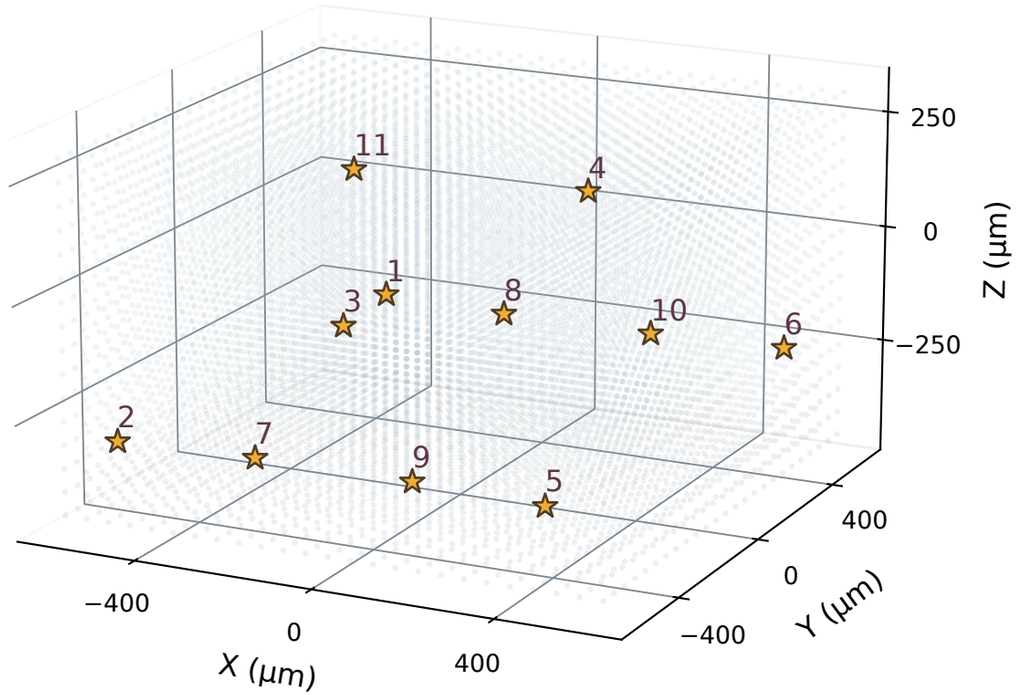
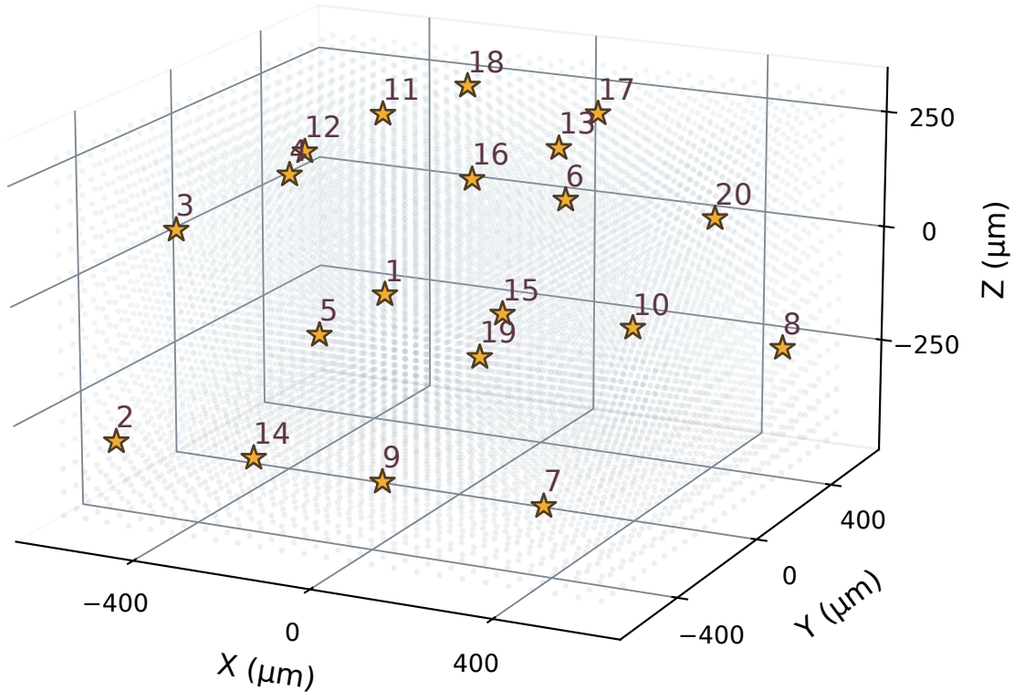
Original reference locations · common camera and physical scale
First bulk onset + 0.1 ms per case; midpoint separation > 301 µm · not an independent continuum-crack count

U · Uniform reference · 11 separated bulk-softening locations



1: bond 55447, t=0.64800 ms; 2: bond 36955, t=0.64850 ms; 3: bond 227013, t=0.67875 ms
4: bond 244315, t=0.68300 ms; 5: bond 396340, t=0.72350 ms; 6: bond 414832, t=0.72350 ms
All additional IDs/times: SPATIAL_DAMAGE_ANALYSIS.json

H · Biphasic · 20 separated bulk-softening locations



1: bond 55447, t=0.55050 ms; 2: bond 36955, t=0.55075 ms; 3: bond 23966, t=0.58250 ms
4: bond 31786, t=0.58500 ms; 5: bond 208097, t=0.58850 ms; 6: bond 225399, t=0.58900 ms
All additional IDs/times: SPATIAL_DAMAGE_ANALYSIS.json